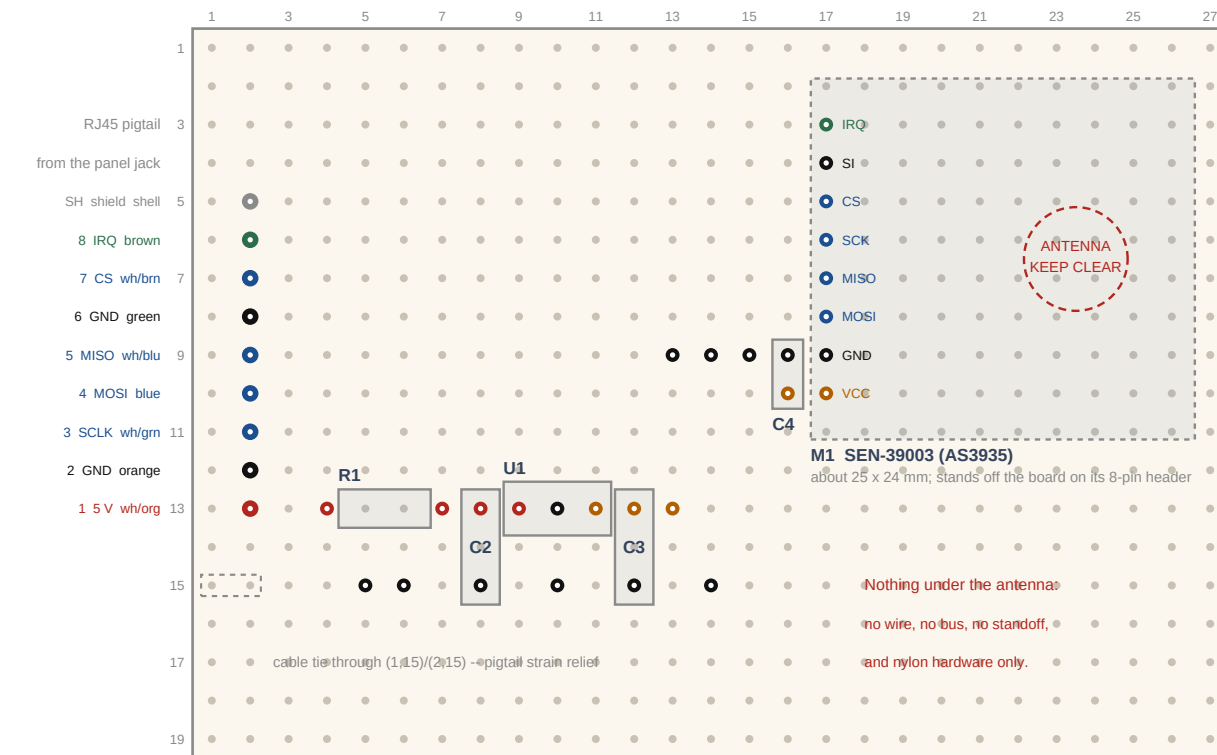


1. SENSOR BOARD -- component placement

5 x 7 cm perf board, isolated pads, viewed from the component side. Everything here is SELV: 5 V and SPI only.



drawn 2:1 -- one grid step = 0.1 in (2.54 mm). Real board 68.6 x 48.3 mm.

5 V 3.3 V GND SPI IRQ no connection

Ref	Part	Holes
R1	100 ohm 1/4 W metal film	(4,13) - (7,13)
C2	47 uF 50 V, EEU-FR1H470	+ (8,13) - (8,15)
U1	MCP1700-3302E, TO-92	VIN (9,13) GND (10,13) VOUT (11,13)
C3	1 uF X7R, C330C105K5R5TA	(12,13) - (12,15)
C4	100 nF X7R, C320C104K5R5TA	(16,10) - (16,9)
M1	SEN-39003 on an 8-pin header	(17,3) .. (17,10), soldered direct

Placement rules that are not negotiable

C4 sits one hole from VCC and one from GND. That tiny loop is the entire point of the part; do not move it to make room.

U1 TO-92, flat face toward you and leads down, is 1 VIN, 2 GND, 3 VOUT. Splay the 0.05 in leads out to 0.1 in.

Hole (10,13) between BUS-A and BUS-B is the LDO's ground pin and is on neither bus. That gap is the input/output isolation.

Confirm which end of the SEN-39003 carries the loop antenna before you solder it, and point that end away from the power chain.

Soldered, not socketed

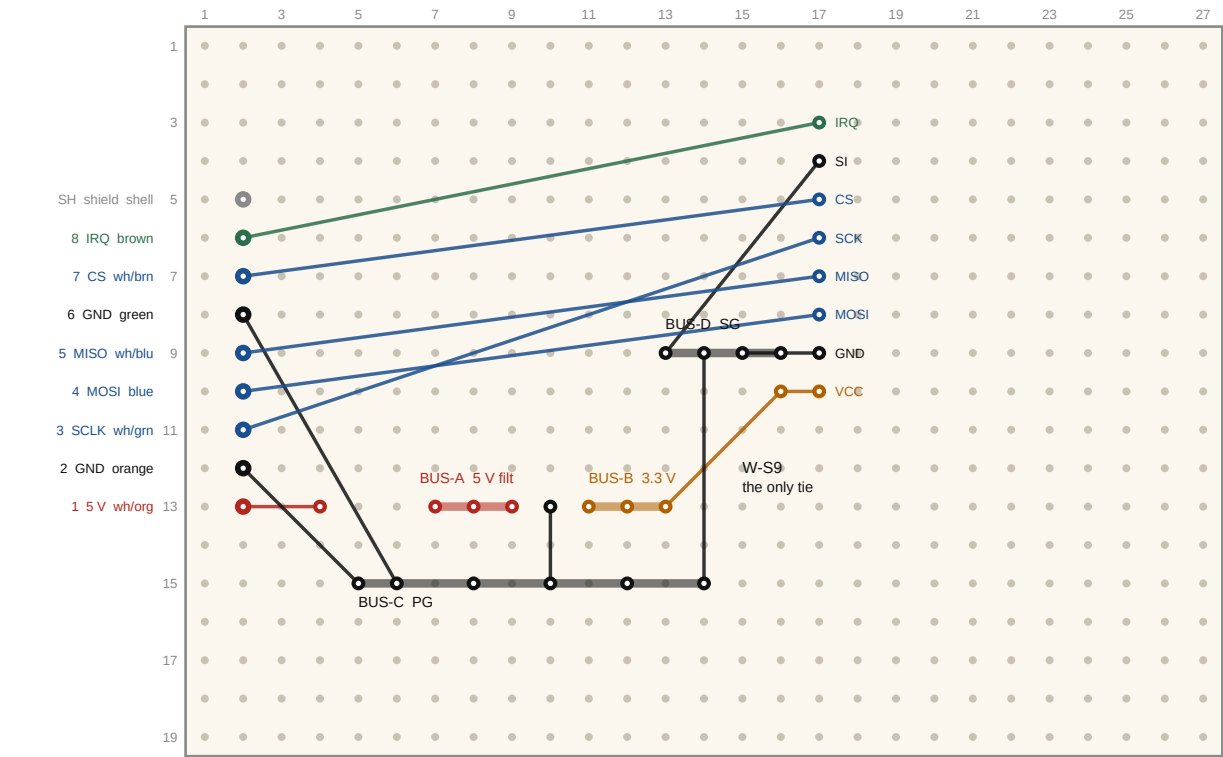
The SEN-39003 header goes straight into the perf. Solderless contacts on this rail are the prime suspect for the step change in README 11.3, and the board is calibrated per unit, so it is not a part you swap casually anyway.

Verify the header order against the silkscreen

The eight pin names above are the order this drawing assumes. Each pin gets its own landing, so if yours differs, only the wire list changes -- nothing moves.

2. SENSOR BOARD -- point-to-point wiring

X-ray view from the component side. All of this is on the BACK of the board, so it mirrors left-right when you flip it over.



drawn 2:1 -- one grid step = 0.1 in (2.54 mm). Real board 68.6 x 48.3 mm.

5 V 3.3 V GND SPI IRQ no connection

Ref	Net	From	To	Wire / note
BUS-A	5 V filtered	row 13	cols 7-9	bare 22 AWG laid across the back of the pads
BUS-B	3.3 V	row 13	cols 11-13	bare 22 AWG laid across the back of the pads
BUS-C	PG power ground	row 15	cols 5-14	bare 22 AWG laid across the back of the pads
BUS-D	SG sensor ground	row 9	cols 13-16	bare 22 AWG laid across the back of the pads
W-S1	5 V in	(2,13)	(4,13)	pin 1 into R1
W-S2	GND pin 2	(2,12)	(5,15)	onto PG
W-S3	GND pin 6	(2,8)	(6,15)	onto PG -- the SCLK return
W-S4	LDO GND	(10,13)	(10,15)	U1 pin 2 down to PG
W-S5	3.3 V out	(13,13)	(16,10)	BUS-B up to the C4 / VCC node
W-S6	VCC link	(16,10)	(17,10)	one hole -- do not lengthen
W-S7	GND link	(17,9)	(15,9)	sensor GND onto SG
W-S8	SI strap	(17,4)	(13,9)	grounds SI: selects SPI, not I2C
W-S9	SG-PG tie	(14,9)	(14,15)	the ONLY tie between the grounds
W-S10	SCLK	(2,11)	(17,6)	
W-S11	MOSI	(2,10)	(17,8)	
W-S12	MISO	(2,9)	(17,7)	
W-S13	CS	(2,7)	(17,5)	
W-S14	IRQ	(2,6)	(17,3)	

Two grounds, one tie

PG carries the cable's ground return, the bulk cap and the LDO reference. SG carries only the sensor's GND pin, its 100 nF and the SI strap. They meet at exactly one place: W-S9. Bridge them anywhere else and you have wrapped a ground loop around the LDO; the 100 nF stops being local and the whole point of the split is gone. Where along PG you tie is not critical -- at 350 uA the drop along the bus is nanovolts. That there is only ONE tie is critical.

C3 references PG, C4 references SG

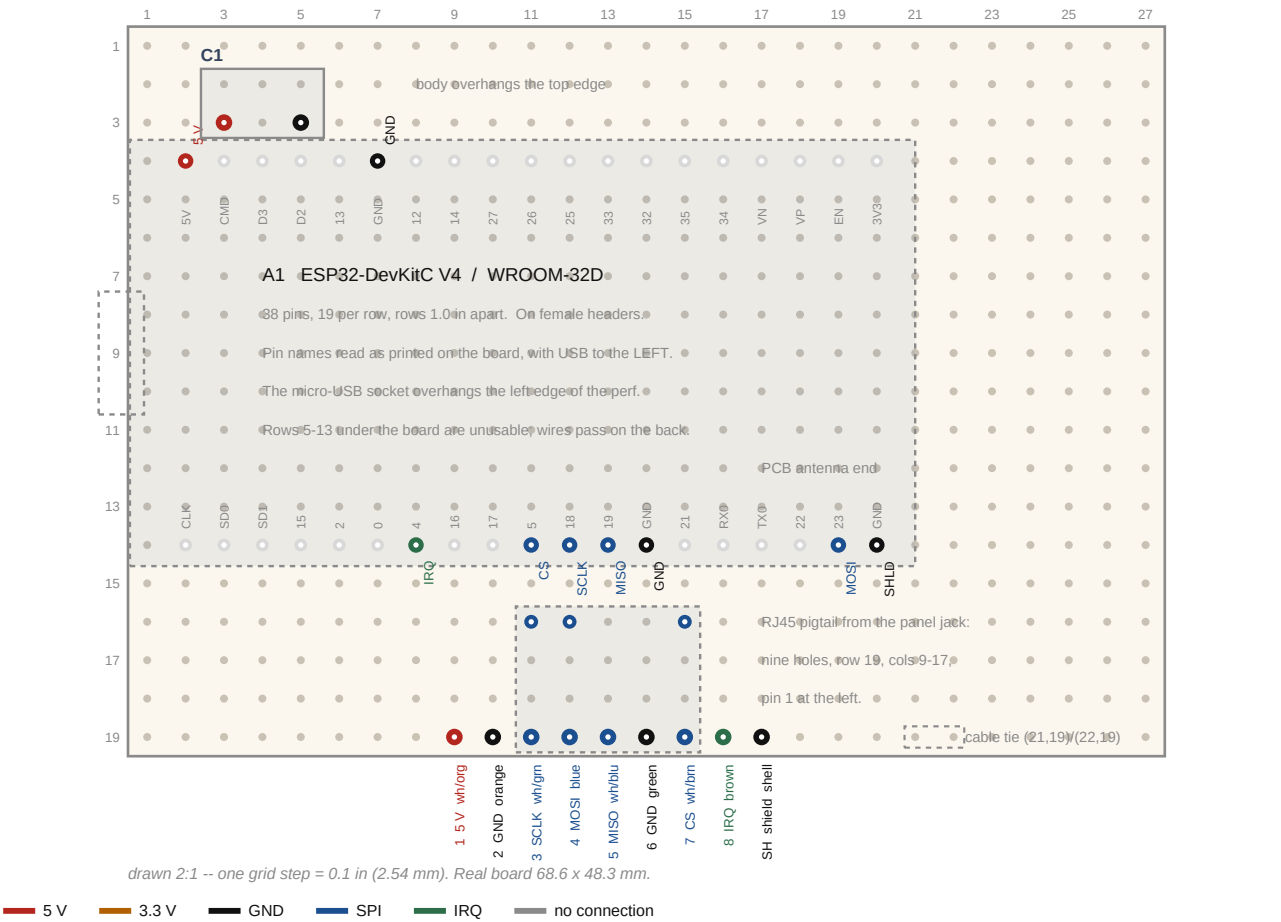
C3 is the MCP1700's stability capacitor, so it belongs to the regulator and lands on PG. C4 is the sensor's decoupling, so it belongs to the sensor and lands on SG. Swapping them defeats the split as surely as a second tie would.

Crossings are fine, except over a bus

Point-to-point links are insulated and run on the solder side; they cross each other freely. The four buses are BARE. Keep every wire clear of them or sleeve it where it passes.

3. MAIN BOARD -- component placement

5 x 7 cm perf board. Contents: the ESP32 dev board, the 5 V bulk cap at its pins, three series positions, and the cable pigtail.



Ref	Part	Holes
A1	ESP32-DevKitC V4 (WROOM-32D)	female headers: row 4 c2-20, row 14 c2-20
C1	470-1000 uF 16-25 V 105 C	+ (3,3) - (5,3) stripe at (5,3)
R2	SCLK series -- wire link	(11,16) - (11,19)
R3	MOSI series -- wire link	(12,16) - (12,19)
R4	CS series -- wire link	(15,16) - (15,19)

Socket this one, unlike the sensor board

Dev boards die, and this one sits metres from the antenna, so the contact-resistance worry that governs the sensor board does not apply. Female headers; keep BOOT and EN reachable.

Check the row spacing before you solder the headers

Drawn at 10 holes (1.0 in), measured off the DevKitC V4. If yours is 0.9 in, ONLY the top header and C1 move: top row becomes row 5, C1 becomes (3,4)/(5,4). Every signal is on the bottom row, which does not move. Use the dev board as its own jig.

C1 cannot be as tight as you want it to be

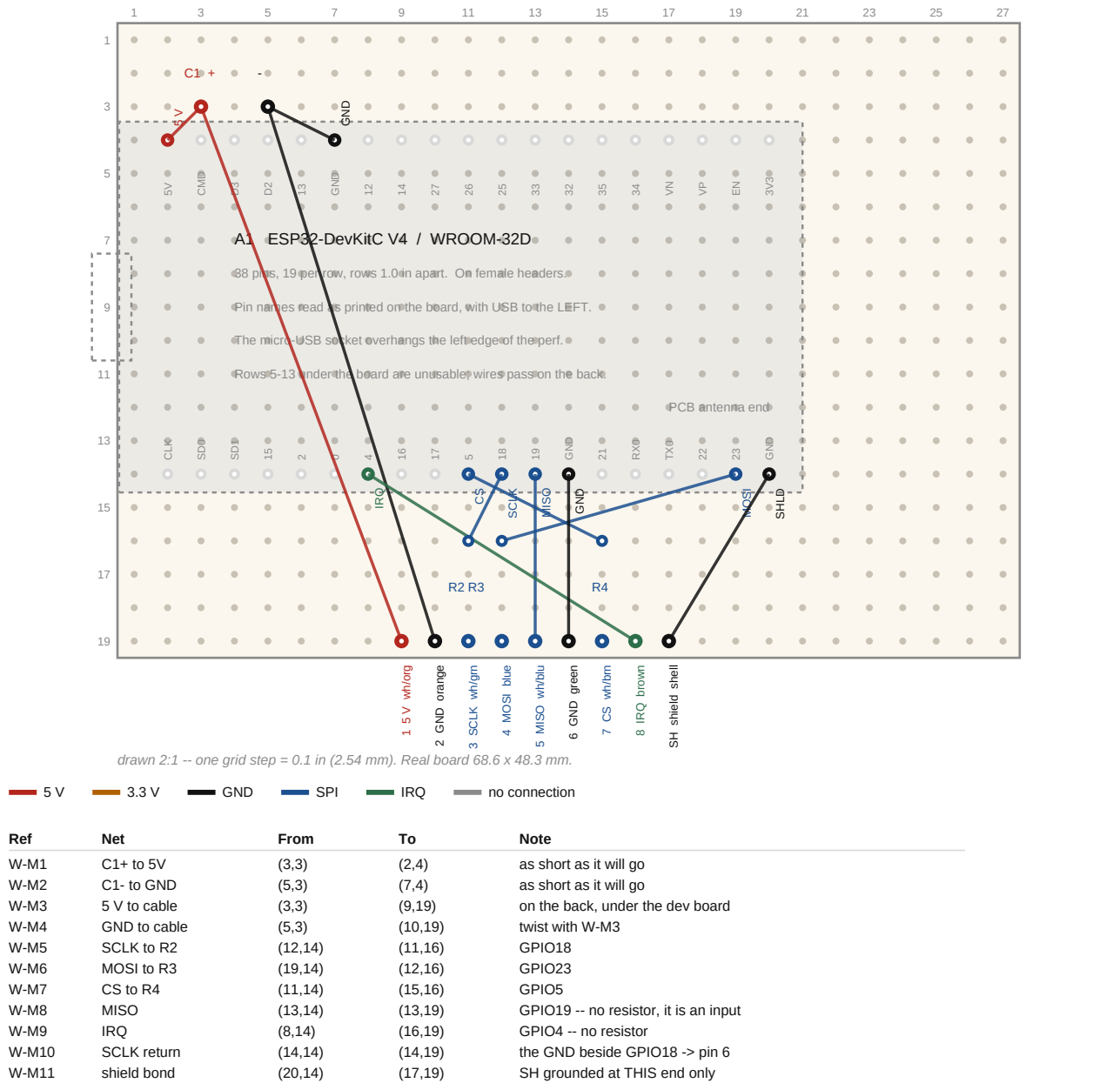
The DevKitC has FIVE pins between 5V and its nearest ground, so the bulk-cap loop is about 27 mm however you arrange it. That is a property of the dev board, not of this layout. Keep W-M1 and W-M2 short and stop optimising.

R2/R3/R4 are wire links on day one

Only the three lines the ESP32 drives get a position. MISO and IRQ are driven from the far end, so damping them here would do nothing. Fit 33-100 ohm only if the 3 m sweep misbehaves.

4. MAIN BOARD -- point-to-point wiring

X-ray view from the component side. The back-side wires run under the dev board, so the two long power runs cost nothing.



C1 is the reason this board exists

470-1000 uF, 105 C, at the 5V and GND pins. It is the reservoir the USB cable cannot supply fast enough during a WiFi burst -- README 7.4. The stripe is the minus lead, at (5,3).

The pair discipline is deliberate

W-M3/W-M4 leave from C1 and land on cable pins 1 and 2, which are a twisted pair. W-M10 takes the GND pin next to GPIO18 to cable pin 6, the other half of the SCLK pair.

The shield is bonded here and nowhere else

W-M11 grounds the jack's SH pin at the main board. At the sensor board SH lands in its hole and stops there. One end only: bond both and a shielded patch cable becomes a ground loop wrapped around the whole run.

Route the USB cable away from the Cat5. Do not bundle them parallel, in the box or outside it.

5. Build order, and the things that will bite

Read this before you cut anything. Sections refer to the project README.

Sensor board

1. Cut/snap the perf to 27 x 19 holes and de-burr. Mark hole (1,1) in a corner with a pen; every coordinate counts from it.
2. Solder the four buses first, while the board is flat and empty: BUS-C PG (row 15, c5-14), BUS-D SG (row 9, c13-16), BUS-A 5 V (row 13, c7-9), BUS-B 3.3 V (row 13, c11-13).
3. R1, C2, U1, C3, C4 -- shortest parts first.
4. W-S9, the single SG-PG tie. Do it deliberately and mark it.
5. Solder the 8-pin header into the SEN-39003, then that assembly into (17,3)..(17,10). Check the antenna end points right, away from the power chain, before you commit.
6. W-S8, the SI strap. Without it the part talks I2C and nothing works at all -- README 5.
7. The five signal links, then the pigtail. Tie at (1,15).
8. Before power: ohmmeter 3.3 V to either ground. Under a few hundred ohms means a bridge -- find it now, not later.

Main board

1. Count your dev board's pins and measure its header row spacing before anything else. Then solder the female headers using the dev board itself as the jig, so the rows end up parallel.
2. C1, stripe at (5,3). Then W-M1 and W-M2, short.
3. R2/R3/R4 as plain wire links. Then the signal wires, then the pigtail. Tie at (21,19).
4. Power it with NO ESP32 in the socket and confirm 5 V appears at cable pin 1 and nowhere it should not.

Things that will bite

The X-ray thing. Both wiring pages are drawn as if you could see through the board from the top, because that is how you place parts. Flip the board to solder and left/right swap. This is the single most common perf-board error.

Isolated pads, not strips. If what you have is stripboard, every row here is already a bus and the layout is wrong without track cuts. Check before you buy.

RJ45 on 0.1 in perf. It does not fit. Panel-mount breakout plus a nine-wire pigtail, both boards -- and wire to the numbers silkscreened on the breakout, never to a position in a photo.

This jack is not Ethernet. It carries 5 V and SPI. A live PoE port puts 48 V on those lines and destroys both ends. Label both boxes now, while you are holding them -- README 7.1.

Nylon standoffs, nylon screws on the sensor board especially. A steel screw beside a 500 kHz loop antenna is a shorted turn.

Solid wire, not the silicone stranded. Buses have to be bare solid or they are not straight; links have to be solid or they will not enter a hole without tinning. README 7.6.

Rigidity is a measurement, not a feeling. README 11.3: the breadboard's noise floor fell by two thirds the moment the build was handled. Phase 2 in README 15 is survey, handle the box, survey again. If the rates move, the build is furniture and nothing measured on it means anything.

Then set data_rate: 200kHz in lightning-detector.yaml before the first survey -- it defaults to 1 MHz, README 7.1.



2.000 in / 50.8 mm -- 20 holes at 0.1 in pitch. If this does not measure true, your printer scaled the page.